

Mohcine BAALLA

Researcher

Tech innovator striving for excellence!

About me

- Selected as a recipient of the **PhD-Associate Scholarship (PASS)**, an excellence-based national program awarded by the CNRST and the Ministry of Higher Education to support high-potential researchers engaged in advanced scientific work, particularly in strategic areas such as artificial intelligence and cybersecurity.
- Passionate about cybersecurity, artificial intelligence, and data science, my work focuses on understanding and securing complex digital systems through data-driven and intelligent approaches. I am particularly interested in the intersection of cybersecurity and AI research, exploring how AI and advanced analytical methods can improve the detection of sophisticated threats in interconnected environments such as IoT systems and large-scale network infrastructures. My goal is to contribute to the development of robust, adaptive security frameworks by studying how intelligent models can identify patterns, anticipate vulnerabilities, and strengthen system resilience.
- Driven by curiosity and a strong analytical mindset, I am deeply interested in investigating the underlying structures that govern data, systems, and digital behaviors. I enjoy approaching complex problems from a research perspective, analyzing patterns, formulating hypotheses, and experimenting with computational models to better understand emerging security challenges. By combining cybersecurity expertise with research in artificial intelligence and data analysis, I aim to contribute to innovative solutions that advance both intelligent systems and the security of modern digital infrastructures.

Education

PhD ENSIAS, Mohammed V University in Rabat, Morocco / Since November 2023

Masters in Systems and Services Security ENSIAS, Mohammed V University in Rabat, Morocco / From September 2021 to September 2023

Bachelor's Degree in Mathematical Sciences and Computer Science Faculté des sciences ben msik Casablanca, Morocco / From September 2020 to July 2021

High School Diploma Mohamed 6 High School Casablanca, Morocco / From September 2017 to July 2018

Compétences techniques

Programming

Python, C, C++, java, Html, Css Javascript, PHP, Script Shell, C#, ExpressJS, Node, JEE, React, Mobile development

Technologies

Artificial Intelligence, Machine Learning, Deep Learning, Data Mining, Data Analysis, Large-Scale Data Processing (Hadoop), Intelligent Systems, Pattern Recognition.

Tools

Nmap, Nessus, Burpsuite, Wireshark, Ettercap, Metasploit, docker, Kubernetes, Fortinet, Cryptography

Environments

Windows, Kali Linux, Ubuntu

Standards

The ISO/IEC 2700x family

DBMS

Oracle, Mysql, SQLite

Virtualization & Hypervisors

VMware vSphere, VirtualBox, Proxmox VE

Simulation

SUMO, NS-3, NS-2, GNS3, Veins, OMNeT++

Version Control

Git, GitHub, GitLab

Skills

Network traffic analysis, Malware Analysis, Incident Response

Project & Issue Tracking

Jira, Confluence, Trello

Adobe

Photoshop, Premiere Pro, After Effect



Languages

- Arabic: **Native**
- French: **Fluent**
- English: **Fluent**

Interests

Scientific news

Video Games

Theater

Football

Discussions and debates

Published Papers

- Baalla, Mohcine, and Bouzidi Driss. *Introducing and Analyzing the Fake Relationship Attack in Social IoT*. In *Proceedings of the Sixth International Conference on Intelligent Computing in Data Sciences (ICDS 2024)*, 23–24 October 2024, Marrakesh, Morocco.
- Baalla, Mohcine, and Bouzidi Driss. *Zero-Knowledge Proofs in Vehicular Networks*. In *Proceedings of the 12th International Conference on Wireless Networks and Mobile Communications*, 25–27 November 2025, Saudi Arabia.
- Baalla, Mohcine, and Bouzidi Driss. *The Hidden Threat: Analyzing Trust Manipulation Attacks in Trust Management Systems for Vehicular Networks*. In *Proceedings of the International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA 2025)*, 7–9 August 2025, Antalya, Turkey.
- Baalla, Mohcine, and Bouzidi Driss. *Machine Learning Detection Approach Against Trust Manipulation Attacks in Trust Management Systems*. In *Proceedings of the Sixth International Conference on Intelligent Computing in Data Sciences (ICDS 2024)*, IEEE, 23–24 October 2024, Marrakesh, Morocco.
- Baalla, Mohcine, and Bouzidi Driss. *New Detection Approach Against Trust Manipulation Attacks in VANETs*. In *Proceedings of the International Conference on Big Data and Internet of Things*, Springer Nature Switzerland, 2024.

Works in Progress

- **Tamper-Proof AI Decision Journaling:** Research on improving transparency and accountability in AI-driven systems. This work proposes a tamper-resistant framework for recording critical AI decision data, including input features, model outputs, confidence scores, and explanation metadata, into an immutable journal. By combining cryptographic hash chaining with distributed ledger mechanisms, the framework ensures the integrity of AI decision logs while enabling reliable forensic analysis and post-incident investigation.
- **Cold Start:** Proposal of a new GNN to assess the trustworthiness of an unknown node in an intelligent network. The goal is to assign a trust value and ensure the initial integration of nodes while maintaining good responsiveness.
- **Centralized vs Decentralized:** Comparative analysis of centralized and decentralized models for trust management. The objective is to find a balance between global control and local resilience, with an exploration of hybrid solutions.

Teaching

As part of my academic engagement at ENSIAS, I have taught several modules combining theoretical foundations and practical applications, with a focus on developing strong analytical and problem-solving skills. My teaching includes courses such as **Linear Programming**, **Graph Theory**, and **Merise Methodology**, where I emphasize modeling, optimization, and structured reasoning.

In addition, I have delivered applied courses including **Mobile Application Development**, **Internet Protocols and Applications**, **Security-Oriented Python Programming**, and **Introduction to Systems Security**. I have also taught **Linux** within the JobInTech program, focusing on system-level understanding and practical competencies.

Through these teaching experiences, I aim to bridge theory and practice while fostering a rigorous approach to computational thinking, data-driven reasoning, and system design—skills that are central to both cybersecurity and artificial intelligence research.

For the upcoming academic period, I am scheduled to teach **Python Programming** and **Cryptography**, further reinforcing my involvement in foundational and advanced topics relevant to secure and intelligent systems.

Research Interests

- Artificial Intelligence Systems and Architectures
- Model Acceleration and Efficient AI (compression, quantization, pruning, and hardware-aware optimization)
- Trustworthy, Explainable, and Transparent AI
- AI Decision Logging, Auditing, and Accountability
- Security, Robustness, and Adversarial Defense in Machine Learning Models
- AI for Cybersecurity, Autonomous Threat Detection, and Network Protection
- Graph Neural Networks and Learning on Networked Data
- Distributed and Decentralized AI Systems
- Data-Centric AI and Representation Learning

Projects

Anomaly Detection in IoT Networks Using Deep Learning
Graph Neural Network for Attack Detection
Development of a Port Scanner (Python)
Detection of DOS/DDOS Attacks Using a Machine Learning Approach (Python)
Conduct In-Depth Research on Google Knowledge Graph
New Trust Model Against TMA Attack (Trust Manipulation Attack)
Detection of Selfish Nodes (Students) in Online Learning Platforms
RFID Card Manipulation (C#)
Credit Risk Classification (Python)
Creation of a Carpooling Web Application (React, Express, PostgreSQL)
Conduct Research on RFID, NFC, and Bluetooth in IoT Systems
Development of a Web Application for Grade Management Using HTML, CSS, jQuery, PHP
Implementation of DSR and AODV Protocols in the NS2 Simulator
Securing a Web Application with JWT, Logging Levels, and TLS/SSL (Python)
Conducting a Case Study on ISO 27003 and ISO 27001
Creation of an IPv6-IPv4 Tunnel Project Using GNS3



@MohcineBAALLA



@MohcineBAALLA